SIMATS ENGINEERING

ASSESSMENT TOOL 1 — PROBLEM STATEMENTS

COURSE NAME: ARTIFICIAL INTELLIGENCE
COURSE CODE: CSA17

ANALYTICAL PROBLEM SOLVING

Problem 1 — Water Jug Problem

You have a 4-gallon jug and a 3-gallon jug, neither marked with measurements. A pump is available to fill either jug with water. Using only filling, emptying (pouring onto the ground), and pouring water from one jug into the other, how can you end up with exactly 2 gallons of water in the 4-gallon jug?

Assumptions: a jug can be filled from the pump; water can be poured out onto the ground; water can be poured from one jug into the other; no other measuring devices are available.

Problem 2 — Mars Rover Agent Analysis

Investigate how a Mars Rover explores and analyzes samples on Mars by answering the following:

i. What are the percepts for this agent? i. Characterize the operating environment. iii. What actions can the agent take? iv. How can the performance of the agent be evaluated? v. What agent architecture is most suitable for this agent?

Problem 3 — 8-Queens Problem

Explore the search space using search strategies and formulate the problem components for the 8-Queens problem: place 8 queens on a chessboard such that none of the queens attack one another. A given configuration where the first-column queen and last-column queen share a diagonal is not a valid solution.

Problem 4 — OLA Cab Booking Agent

A customer wants to travel from one location to another using the OLA Cab booking mobile application, choosing between cab types such as Mini, Micro, Sedan, Shared, Prime, etc., based on comfort. Identify the type of problem-solving agent used and write the pseudocode for this scenario.

Problem 5 — Logistics Routing with Uniform Cost Search

A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). Determine the least-cost path from a starting warehouse S to a destination warehouse G using the Uniform Cost Search (UCS) algorithm. The delivery network is shown below: